Applications: Nuclei and Accelerators

Four problems, one ledger

SYNOPSIS

Everything in this module so far has been **forces and fields**: how hard charges push, and what the push per coulomb is at a given place. This note spends that machinery on **four real objects** — a nucleus coming apart, two nuclei being driven together, and a beam of particles crossing a pair of charged plates.

There is **no new physics in it at all**. Every one of the four is the same bookkeeping line, written out and then crossed down:

$$U_1 + K_1 + W_{\text{ext}} = U_2 + K_2$$

Nothing external does work in any of them, so W_{ext} goes. In each problem two more terms are zero because something starts at rest, or ends at rest, or is far enough away that its potential energy is nothing. **What survives is a one-line equation**, and the whole skill is knowing which terms to cross out.

The two expressions for U you already have:

$U = kq_1q_2 / r$ (two point charges) $U = q\Delta V$ (a charge crossing a potential difference)

WHERE THE OTHER APPLICATIONS ARE

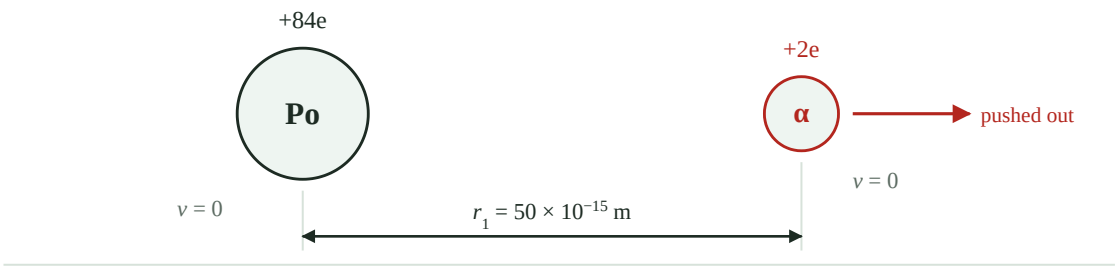
Two applications you might expect here are already elsewhere in this module, and are not repeated. **The electric force against gravity** is Note 8 — the whole isomorphism table, plus his ladder sheet with the 2 C charge beside the 2 kg mass. **Millikan** is Note 9, where the oil drop hangs motionless and $qE = mg$ gives 4.8×10^{-19} C, and the method of finding a quantum from measurements alone is the five containers of pennies in *Charged Objects*.

PERFORMANCE EXPECTATION

1. **Solve** problems involving electric potential energy, using conservation of energy to find speeds, closest approaches and accelerating voltages for charged particles.

1. Alpha decay: where does the speed come from?

A radon nucleus (atomic number 86) transforms spontaneously into a polonium nucleus (atomic number 84) and an **alpha particle** — two protons and two neutrons bound together. Immediately afterwards both pieces are **at rest**, about 50×10^{-15} m apart.



Everything the alpha ends up with is sitting here as potential energy.

Figure 1. His sketch of the instant after the transformation. The polonium carries +84*e*, the alpha +2*e*, and **neither is moving**. Every joule the alpha ends up with is already in the diagram, stored as potential energy in that separation.

(a) The potential energy it starts with

Work it in two steps, the way he does: first the **potential** the polonium creates at the alpha's position, then the **energy** that potential gives to the alpha's charge.

(b) How fast the alpha is going when it is clear

Far away, the potential energy is gone. The polonium is about fifty times heavier than the alpha, so it barely moves, and **the alpha takes essentially all of the kinetic energy**.

(c) Where is it when it has 90% of that speed?

This is the part worth slowing down on, and it is also the one place on his sheet where the arithmetic goes wrong. The **set-up is right**:

$$U_1 = \tfrac{1}{2}m_\alpha(0.9v)^2 + kq_{Po}q_\alpha / r'$$

$$\rightarrow \quad r' = kq_{Po}q_\alpha / [\, U_1 - \tfrac{1}{2}m_\alpha(0.9v)^2 \,]$$

The denominator is **what is left over as potential energy** — the total minus the kinetic. Before substituting anything, notice that you can read it off without a calculator.

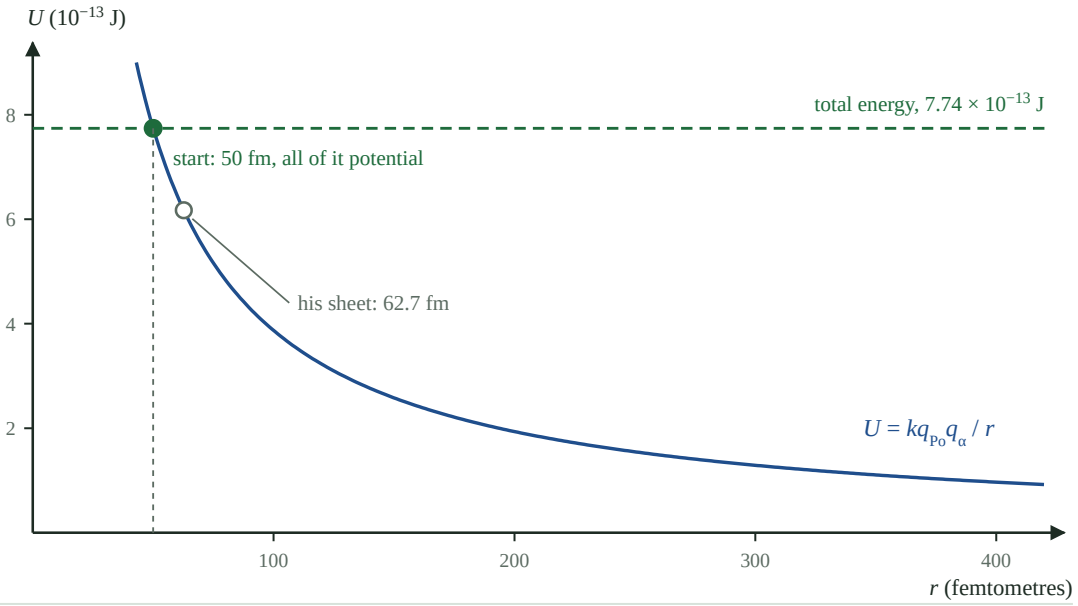


Figure 2. The curve, the total-energy line and the start are drawn for you, and so is the mark where his sheet's answer falls. **Put the 90% point on it:** show the kinetic energy as a bar from the dashed line down to the curve, and read off *r*.

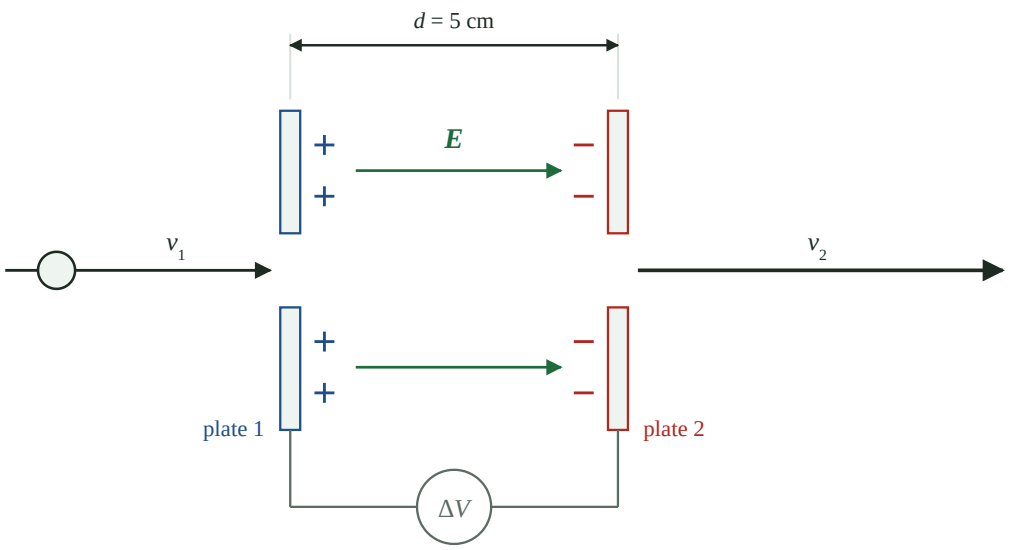
(b) Fire the deuterium at the protons instead

His part (b) swaps the roles. **Nothing in the physics changes** — the barrier is the same height, because it depends on the two charges and the distance, and neither has moved. Only the mass in the kinetic-energy term is different.

(c) Launch at half the speed — how close now?

3. The accelerator: plates with holes in them

The last two problems are the same ledger with the **other** expression for U. Between parallel plates there is no $1/r$ anywhere — the field is uniform, and the energy a charge gains is simply $q\Delta V$.



The holes are what make it an accelerator rather than a capacitor: the beam goes in one side and out the other, and the only thing that changed is its kinetic energy.

Figure 4. His sketch of the device. Two oppositely charged plates 5 cm apart, with **small holes cut in them** so a beam can pass straight through. The particle enters at v_1 and leaves at v_2; the only thing that happened in between is that the field did work on it.

(a) An electron speeds up — what is the voltage?

An electron is injected at 2.0×10^7 m/s and leaves at 2.5×10^7 m/s. **Run the ledger the other way round**: the speeds are given and the potential difference is the unknown.

(b) Now send a proton through the same device

Same plates, same 640.5 V, same entry speed. **Before calculating, predict.**

(c) The field between the plates

The field follows from the voltage and the gap, and his line for it is worth copying out because it shows **where *V = Ed* comes from** rather than just using it:

$$V_{12} = V_2 - V_1 = -\int \mathbf{E} \cdot \mathrm{d}\mathbf{r} = -\int E \, \mathrm{d}r \cos(180°)$$

$$= E \int \mathrm{d}r = Er$$

Walking *against* the field is what makes the cosine −1, and the two minus signs cancel. With a **uniform** field E comes out of the integral and the rest is just the gap width.

4. A 20 kV accelerator, and the question about the gap

His fourth problem is the same device driven much harder: a proton injected at 2.0×10^6 m/s across **20 kV**, with the left plate at the higher potential and the plates 5 cm apart.

Now widen the gap to 10 cm, holding 20 kV

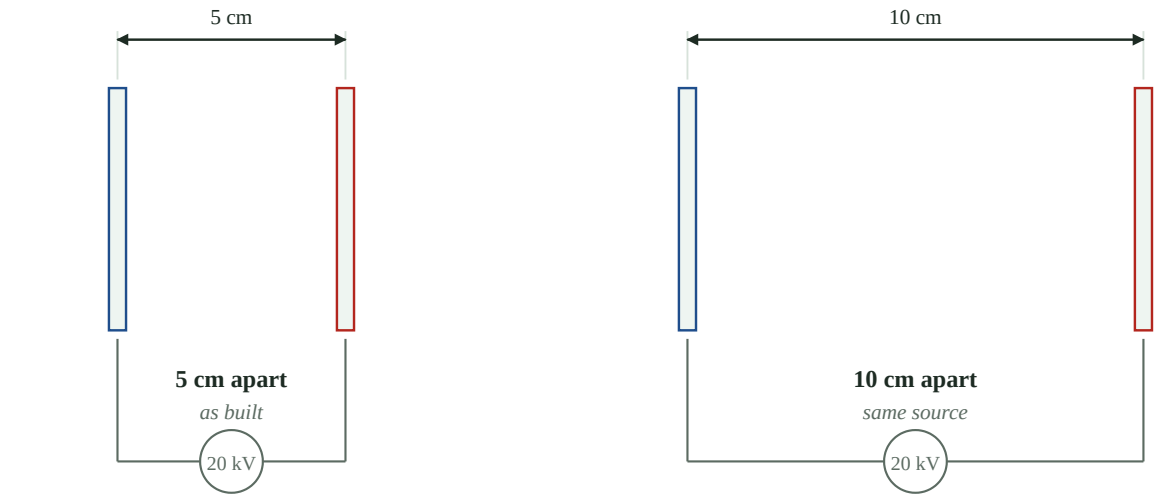
His answer to this one is written out in prose, and it is the best line on the sheet:

> **IN HIS WORDS**
>
> “Change in separation of the plates doesn’t come into play given the equation for v_2 above.”

No separation appears in it. The energy a charge picks up is $q\Delta V$, and that is the whole story — the proton leaves at the same 2.80×10^6 m/s whether the plates are 5 cm or 10 cm apart. It takes longer to cross, and the force on it is

weaker, and those two exactly compensate.

But **something** must have changed. What?



Given: the gap on the right is twice the gap on the left, and the same source sits across both.
Mark the field between each pair and the charge on each plate, and say how the two sides differ.

Figure 5. Mark the field between each pair and the charge on each plate. Same source, twice the gap on the right — what is the same for the beam, and what is not?

Your job on Figure 5. Mark the field between each pair and the charge sitting on each plate. Then say in one sentence what is the same on the two sides and what is not — and which of the two the beam notices.

Looking ahead

That is the electricity module finished. Everything in it came from two equations for the force and two for the energy, and the last four problems were those same equations handed to a nucleus and a particle beam.

Notice what the accelerator could **not** do. It changed the beam's speed and left its direction alone — the holes had to line up, because the particle goes where it was already going. **An electric field cannot steer.**

Watch for

The next module is **magnetism**, and it starts exactly where this one stops: a particle is accelerated through a potential difference, and then it is **bent**. On the same sheet these four problems came from, the next two pages do that — a proton accelerated through ΔV into a field B, turning on a circle of radius $R = (1/B)\sqrt{(2m\Delta V/q)}$.

Everything you just did stays: that radius has your ΔV inside it, and the accelerator is still the thing that sets the speed.